Close

Sohaa Jamadar | 1252090024 | 24 | SEDA-B1 | Assignment 10

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```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
```

Load Dataset

```
df = pd.read_csv("temperatures.csv")
df
```

	YEAR	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	ANNUAL	JAN-FEB	MAR-MAY	JUN-SEP	OCT-DEC
0	1901	22.40	24.14	29.07	31.91	33.41	33.18	31.21	30.39	30.47	29.97	27.31	24.49	28.96	23.27	31.46	31.27	27.25
1	1902	24.93	26.58	29.77	31.78	33.73	32.91	30.92	30.73	29.80	29.12	26.31	24.04	29.22	25.75	31.76	31.09	26.49
2	1903	23.44	25.03	27.83	31.39	32.91	33.00	31.34	29.98	29.85	29.04	26.08	23.65	28.47	24.24	30.71	30.92	26.26
3	1904	22.50	24.73	28.21	32.02	32.64	32.07	30.36	30.09	30.04	29.20	26.36	23.63	28.49	23.62	30.95	30.66	26.40
4	1905	22.00	22.83	26.68	30.01	33.32	33.25	31.44	30.68	30.12	30.67	27.52	23.82	28.30	22.25	30.00	31.33	26.57
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
112	2013	24.56	26.59	30.62	32.66	34.46	32.44	31.07	30.76	31.04	30.27	27.83	25.37	29.81	25.58	32.58	31.33	27.83
113	2014	23.83	25.97	28.95	32.74	33.77	34.15	31.85	31.32	30.68	30.29	28.05	25.08	29.72	24.90	31.82	32.00	27.81
114	2015	24.58	26.89	29.07	31.87	34.09	32.48	31.88	31.52	31.55	31.04	28.10	25.67	29.90	25.74	31.68	31.87	28.27
115	2016	26.94	29.72	32.62	35.38	35.72	34.03	31.64	31.79	31.66	31.98	30.11	28.01	31.63	28.33	34.57	32.28	30.03
116	2017	26.45	29.46	31.60	34.95	35.84	33.82	31.88	31.72	32.22	32.29	29.60	27.18	31.42	27.95	34.13	32.41	29.69

Next steps: Generate code with df New interactive sheet

Data Info.

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 117 entries, 0 to 116
Data columns (total 18 columns):
 #   Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0   YEAR     117 non-null    int64
 1   JAN      117 non-null    float64
 2   FEB      117 non-null    float64
 3   MAR      117 non-null    float64
 4   APR      117 non-null    float64
 5   MAY      117 non-null    float64
 6   JUN      117 non-null    float64
 7   JUL      117 non-null    float64
 8   AUG      117 non-null    float64
 9   SEP      117 non-null    float64
 10  OCT      117 non-null    float64
 11  NOV      117 non-null    float64
 12  DEC      117 non-null    float64
 13  ANNUAL   117 non-null    float64
 14  JAN-FEB  117 non-null    float64
 15  MAR-MAY  117 non-null    float64
 16  JUN-SEP  117 non-null    float64
 17  OCT-DEC  117 non-null    float64
dtypes: float64(17), int64(1)
memory usage: 16.6 KB
```

```
df.describe()
```

	YEAR	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	
count	117.000000	117.000000	117.000000	117.000000	117.000000	117.000000	117.000000	117.000000	117.000000	117.000000	11
mean	1959.000000	23.687436	25.597863	29.085983	31.975812	33.565299	32.774274	31.035897	30.507692	30.486752	2
std	33.919021	0.834588	1.150757	1.068451	0.889478	0.724905	0.633132	0.468818	0.476312	0.544295	
min	1901.000000	22.000000	22.830000	26.680000	30.010000	31.930000	31.100000	29.760000	29.310000	29.070000	2
25%	1930.000000	23.100000	24.780000	28.370000	31.460000	33.110000	32.340000	30.740000	30.180000	30.120000	2
50%	1959.000000	23.680000	25.480000	29.040000	31.950000	33.510000	32.730000	31.000000	30.540000	30.520000	2
75%	1988.000000	24.180000	26.310000	29.610000	32.420000	34.030000	33.180000	31.330000	30.760000	30.810000	3
max	2017.000000	26.940000	29.720000	32.620000	35.380000	35.840000	34.480000	32.760000	31.840000	32.220000	3

```
df.shape
```

```
(117, 18)
```

Implement Data Preprocessing (No cleaning required as data is already clean)

Converting wide format to long format

```
df1 = df.melt(
    id_vars=['YEAR'],
    value_vars=['JAN','FEB','MAR','APR','MAY','JUN','JUL','AUG','SEP','OCT','NOV','DEC'],
    var_name='Month',
    value_name='Temperature'
)

df1
```

	YEAR	Month	Temperature
0	1901	JAN	22.40
1	1902	JAN	24.93
2	1903	JAN	23.44
3	1904	JAN	22.50
4	1905	JAN	22.00
...	...	...	...
1399	2013	DEC	25.37
1400	2014	DEC	25.08
1401	2015	DEC	25.67
1402	2016	DEC	28.01
1403	2017	DEC	27.18

1404 rows × 3 columns

Next steps: Generate code with df1 New interactive sheet

Converting month to numeric value

```
df1['Month'] = pd.to_datetime(df1['Month'], format='%b').dt.month
df1
```

	YEAR	Month	Temperature
0	1901	1	22.40
1	1902	1	24.93
2	1903	1	23.44
3	1904	1	22.50
4	1905	1	22.00
...	...	...	...
1399	2013	12	25.37
1400	2014	12	25.08
1401	2015	12	25.67
1402	2016	12	28.01
1403	2017	12	27.18

1404 rows × 3 columns

Next steps: Generate code with df1 New interactive sheet

Define features

```
X = df1[['Month']]
y = df1['Temperature']
```

Data validation

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

Linear Regression

```
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Prediction = ", y_pred)
```

```
Prediction =  [29.283892   29.08214986 29.12249829 29.12249829 29.24354357 29.00145301
 29.283892   28.92075615 28.92075615 29.16284672 29.00145301 29.12249829
 28.92075615 29.16284672 29.08214986 29.08214986 29.20319514 28.92075615
 29.04180143 29.36458885 29.08214986 29.283892   29.283892   29.20319514
 29.12249829 29.16284672 29.08214986 28.92075615 28.96110458 29.24354357
 29.24354357 28.92075615 28.92075615 29.283892   28.92075615 29.24354357
 29.283892   29.08214986 29.04180143 29.20319514 29.12249829 28.96110458
 29.32424042 28.96110458 29.16284672 29.32424042 29.04180143 29.16284672
 29.00145301 29.00145301 29.12249829 29.24354357 29.12249829 28.96110458
 29.283892   29.04180143 28.92075615 29.24354357 29.00145301 29.283892
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 29.24354357 29.12249829 28.92075615 28.96110458 29.24354357 28.96110458
 29.283892   28.96110458 29.08214986 29.00145301 29.12249829 29.12249829
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 29.04180143 29.00145301 29.16284672 29.36458885 29.16284672 29.20319514
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 29.16284672 29.12249829 29.16284672 29.16284672 29.04180143 29.08214986
 28.92075615 29.24354357 29.283892   29.24354357 29.36458885 29.00145301
 29.283892   29.04180143 29.32424042 29.08214986 29.08214986 29.16284672
 29.24354357 29.20319514 28.96110458 29.283892   29.08214986 29.283892
 29.12249829 28.96110458 29.20319514 29.16284672 29.04180143 29.04180143
 29.283892   29.36458885 29.16284672 29.08214986 29.16284672 29.32424042
 29.08214986 29.08214986 29.08214986 29.32424042 28.92075615 29.36458885
 29.36458885 29.32424042 28.92075615 28.92075615 29.16284672 29.16284672
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 29.32424042 28.96110458 29.36458885 28.92075615 29.36458885 29.283892
 29.283892   29.16284672 29.04180143 28.92075615 29.32424042 29.20319514
 29.20319514 29.283892   28.96110458 29.283892   29.08214986 28.96110458
 29.00145301 29.24354357 29.08214986 29.04180143 29.36458885 29.283892
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 29.36458885 29.12249829 29.24354357 29.00145301 29.32424042 29.08214986
 29.32424042 29.24354357 29.16284672 29.12249829 29.32424042 29.20319514
 29.12249829 29.04180143 28.92075615 29.16284672 29.36458885 29.12249829
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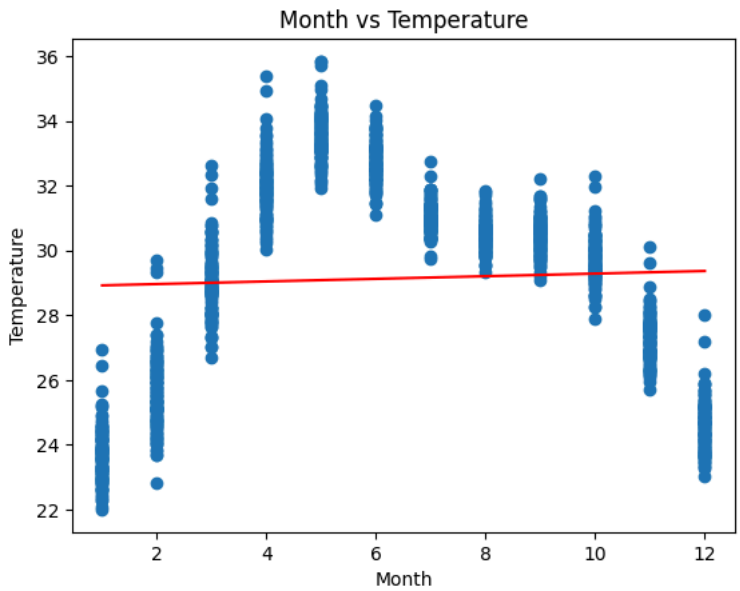
```
 29.36458885 29.32424042 29.04180143 29.16284672 29.00145301 28.92075615
 29.36458885 28.96110458 29.36458885 29.283892   29.00145301 29.00145301
 29.16284672 29.00145301 29.36458885 29.12249829 29.08214986 29.04180143
 29.12249829 28.96110458 28.92075615 29.32424042 29.283892   28.96110458
 29.36458885 29.00145301 29.283892   29.20319514 29.20319514 29.36458885
 28.92075615 28.96110458 29.20319514 29.00145301 29.16284672 29.08214986
 29.32424042 28.96110458 29.32424042 29.32424042 29.04180143]
```

Evaluation metrics

```
print("Mean Squared Error:", mean_squared_error(y_test, y_pred))
print("R2 Score: ", r2_score(y_test, y_pred))
print("Root Mean Squared Error", np.sqrt(mean_squared_error(y_test, y_pred)))
print("Mean Absolute Error", mean_absolute_error(y_test, y_pred))
```

```
Mean Squared Error: 9.999806682283973
R2 Score:  -0.005573628993982727
Root Mean Squared Error 3.162247093805918
Mean Absolute Error 2.705271869781447
```

Visualization

```
plt.scatter(X, y)
plt.plot(X, model.predict(X), color="red")

plt.xlabel("Month")
plt.ylabel("Temperature")
plt.title("Month vs Temperature")

plt.show()
```